

STUDENT	
Student number	s3693414
Last name(s)	Xiong
First name(s)	Xiaoyan
Study field	(MSc) Computer Science
Study Level	MSc
Faculty	Faculty of Electrical Engineering, Mathematics and Computer Science
Research chair	Pervasive Systems
Confidential	
Start date assignment	25/02/2026
Expected end date assignment	25/08/2026

GRADUATION COMMITTEE	
1. Committee chair	Durmaz - Incel, Özlem, dr. PhD, PhD
2. Committee member (UT)	Le Viet Duc, Duc, dr.
3. External member	dr. Suzan Bayhan
4. Additional member(s)	Janarthanan Sundaram, ,

ASSIGNMENT DESCRIPTION	
Title of the assignment	A Hierarchical Time-Aware Transformer for Flow-Level Network Intrusion Detection
Assignment description	This assignment proposes a novel, hierarchical Transformer-based framework for Network Intrusion Detection Systems (NIDS) designed to operate effectively in encrypted environments. Unlike traditional methods that rely on aggregated flow-level statistics and discard sequential context, this approach models network traffic by treating bidirectional flows as ordered packet sequences without relying on payload content. The framework is structured into two main stages: a time-aware intra-flow Transformer that captures continuous-time dynamics and packet-level dependencies, followed by an inter-flow Transformer that analyzes contextual relationships across neighboring flows. By utilizing a focal loss objective to mitigate severe class imbalances and leveraging both public (CIC-IDS2017) and internal PCAP datasets, the assignment aims to deliver a robust, representation-learning network detector.
Additional Remarks	

COURSE CODE(S)	
Course code 1	192199978
Course code 2 (if appl.)	
Course code 3 (if appl.)	
Course code 4 (if appl.)	

RECEIVING ORGANISATION (if applicable)	
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CONFIDENTIALITY	
Proposal form confidential	☐
If yes: confidential until	,

SIGNATURES	
Student: Name: Xiaoyan Xiong 01/07/2026 (date (dd-mm-yyyy)) <i>Xiaoyan Xiong</i> (Signature)	Chairman: Name: Durmaz - Incel, Özlem, dr. PhD, PhD (date (dd-mm-yyyy)) (Signature) OPTIONAL 2nd Supervisor Name: Le Viet Duc, Duc, dr. (date (dd-mm-yyyy)) (Signature)